HS Data Science 26

Session 02: Scientific Working I — The Research Loop

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21 April 2026

Session Overview

Module A — The Research Loop

Steps 1–11 of the scientific cycle:

- from reading literature to communicating results
- hypothesis as the contract with the reader
- three experiment modes: dev loop · pre-experiment · full

Module B — Experiment Design

- conditions, controls, and ablations
- choosing and applying the right metric
- pitfalls that invalidate results

Why this session?

Your seminar project is a *mini research loop.* Every step applies — even at small scale.

> **The key insight:** A well-designed experiment with a modest model beats a sophisticated model with a broken experiment every time.

The Research Loop

> **Guiding Question:** What steps turn a vague question into a publishable result?

The Research Loop — Overview

The Research Loop — 11 numbered steps

The Scientific Research Loop as a cycle diagram

Step 1 — Reading Literature

The three-pass method (Keshav 2007)

Pass	What you read	Time
1st	title, abstract, conclusion	5 min
2nd	intro, headings, figures	20 min
3rd	full paper, related work	60–90 min

Only go to pass 3 for **directly relevant** papers.

While reading, ask:

- What claim does this paper make?
- What evidence supports it?
- What does it assume or leave out?
- What would a follow-up study do?

Active reading — example questions

- *"Could this result be a confound?"*
- *"What changes if the dataset is different?"*
- *"Does this generalise beyond the benchmark?"*

Keep a reading log: one sentence per paper — the claim and why you care. Use Obsidian or Notion. Build this habit now.

> Reading is not collecting — it is **understanding what is settled** and **what remains open.**

Step 2 — Finding the Gap

Types of gap *(be complete — gaps fall into these categories)*

Type	Example signal
Missing condition	"Only tested on English"
Conflicting results	"Papers A and B disagree"
Missing baseline	"No comparison to simple heuristic"
Scope limitation	"Offline only, not online"
Scalability limit	"Fails above 1M documents"
Methodological flaw	"No significance test"

Where to look:

- "Limitations" and "future work" sections

A gap is not "nobody did this".
A gap is a *tension, a limit, or an open question that matters* — and that is investigable within your constraints.
From gap to research question:

> *"Does query reformulation improve answer quality in agentic search on the Open Web Index?"*

The gap drives the question. The question drives the hypothesis. The hypothesis drives everything else.

- hedging language: *"we suspect", "future work should"*
- points of disagreement across papers

> Ask: *"What would a practitioner need that the literature does not yet provide?"*

Step 3 — The Research Question

From vague to sharp

Too vague	Sharp
"Is RAG better?"	"Does BM25 re-ranking improve answer faithfulness on TriviaQA vs. dense-only RAG?"
"Does my system work?"	"Does my system outperform a keyword-search baseline on OWI Curlie using nDCG@10?"

A sharp question names: **method · dataset · metric · comparison**.

Criteria for a good research question

- **Specific** — names method, dataset, and metric
- **Empirical** — requires evidence, not just reasoning
- **Interesting** — the answer changes how someone acts
- **Testable** — we can design an experiment to answer it
- **Feasible** — we have the resources and time to run it

> **Testable ≠ Feasible.** *Testable*: can we construct the right experiment? *Feasible*: do we have the compute, data, and time? Both must be true.

Step 4 — The Hypothesis

Step 5 — Experiment Design

Design before you code

Element	Decision to make
Conditions	which system variants to compare
Baseline	simplest reasonable alternative
Dataset	which data, which split, held-out test
Metric	primary + one secondary
Protocol	how many runs, how to average
Resources	GPU-hours, storage, API cost budget

Check resources before committing. A full experiment that exceeds your compute

Pre-registration

Write down hypothesis + design *before* seeing results. This prevents:

- p-hacking
- HARKing (Hypothesising After Results are Known)
- post-hoc cherry-picking

Minimum for this seminar

1. One strong baseline
2. One primary condition
3. One ablation
4. Fixed train / dev / test split — no leakage
5. ≥ 3 runs with different seeds

A one-paragraph design document written before running is worth more than any metric budget is a wasted design.

improvement.

Steps 6 & 7 — Three Experiment Modes

Three Experiment Modes in AI/ML Research

Step 6 — Implementation

The development loop mindset

Work in small, verifiable steps:

1. code → run → inspect output
2. assert shapes, types, and ranges
3. overfit one batch deliberately — if you can't, the pipeline is broken
4. commit working state to Git after each milestone

Version control is not optional

```
feat: BM25 baseline on MSMARCO dev

MRR@10 = 0.187 — matches Thakur et al. 2021.
Pipeline confirmed correct. Next: dense retrieval.
```

Environment and reproducibility

- pin dependencies (`requirements.txt` or `pyproject.toml`)
- fix random seeds at the top of every script
- keep data preprocessing in code — never edit raw data manually
- log every hyperparameter to a config file, not inline

Sanity checks before scaling up:

- Does random perform worse than your model?
- Does overfitting one batch produce near-zero loss?
- Do evaluation scores on dev match your expectation from related work?

Step 7 — Running Experiments · Step 8 — Results

Running experiments systematically

- one experiment = one folder with `config.yaml` + `results.json` + `README.md`
- log **all** runs — including the ones you will not report
- run every condition with ≥ 3 seeds
- measure wall-clock time and GPU memory — part of your resource budget

From raw numbers to reported results

1. Compute mean ± std across seeds
2. Compare conditions on the **held-out test set — once**
3. Do not re-run on test after seeing results
4. Plot distributions, not just means

Honest reporting checklist

- [ ] All conditions reported, not just the best
- [ ] Held-out test used only once
- [ ] Variance reported (not just mean)
- [ ] Baseline is as strong as you could make it
- [ ] No data leakage in the split
- [ ] Statistical significance tested

> **A smaller honest result is more valuable than a large result that cannot be reproduced.**

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Step 9 — Ruling Out Errors

Common error types

Error	What it looks like
Data leakage	test items in training context
Wrong metric	metric ≠ what the hypothesis claims
Evaluation bug	off-by-one, wrong split, wrong average
Confound	model memorises dataset artefacts
Weak baseline	unfair comparison
Random variance	one lucky seed drives the result

Debugging process

1. Sanity-check: does random perform worse?
2. Overfit one batch on purpose
3. Inspect 20–30 model outputs manually
4. Ablate one component at a time
5. Re-run with 3 different seeds
6. Trace train/dev/test construction end-to-end

> **If results are surprisingly good:** Stop and verify. Unexpectedly good results are more suspicious than bad ones.

> Rule out errors *before* writing up — not during the

Error	What it looks like
Distribution shift	train/test from different sources

reviewer response.

Source: *Pineau et al. 2021; Dodge et al. 2019*

Step 9b — Limitations

What a limitation is

A limitation is a boundary on what your results can claim — not an excuse, not a weakness to hide. Every study has them. Naming them honestly is a mark of scientific maturity and protects you from overreach.

Types of limitations

Type	Example
Dataset scope	Evaluated on English web text only — findings may not transfer to other languages or domains
Evaluation protocol	Automatic metrics (BLEU, NDCG) may not align with human judgement
Threat to validity	Participants were self-selected; results may not generalise to the full population

How to write a limitations section

1. **Be specific** — "our dataset may not be representative" is not enough. State *which* aspect is limited and *why* it matters for your conclusions.
2. **Connect to claims** — link each limitation to the research question it affects. Does it change the direction of your finding, or only its magnitude?
3. **Do not apologise** — limitations frame the scope of validity; they are not confessions.
4. **Distinguish limitations from future work** — a limitation constrains the current result; future work is what you would do to lift the constraint.

> **In your seminar narrative:** identify at least one concrete limitation of your planned approach *before* you run any experiments.

Type	Example
Computational budget	Only one model size tested; scaling behaviour unknown
Construct validity	"Relevance" is operationalised as click-through — this captures one dimension of relevance, not all

> Early acknowledgement prevents overreach in the final report.

Steps 10–11 — Write-up and Communicate

The paper is an argument, not a report

Section	Role
Abstract	claim + evidence + impact in 150 words
Introduction	motivation + gap + your contribution
Related Work	what exists and why it falls short
Method	what you did and why
Experiments	how you tested it
Results	what the evidence shows
Analysis	what it means and its limits
Conclusion	restate claim + future work

Write the abstract last — once you know what you found.

Consolidation checklist

- ☐ One central claim — everything supports it
- ☐ Every figure caption states the takeaway
- ☐ Limitations section is honest
- ☐ Code and config available for reproduction

Communication formats

- **Talk:** 12-minute argument — one key idea
- **Poster:** one question, one answer, one figure
- **Report:** ACM format, max 8 pages

> Say one thing clearly. Scientists remember papers that make a single sharp point.

Experiment Design in AI/ML

> **Guiding Question:** How do you design an experiment that actually answers your research question?

Experiment Design — Overview

Topics in this module

- conditions, controls, and ablation studies
- evaluation metrics — choosing and applying them correctly
- common pitfalls that invalidate results
- statistical significance and effect size

Practical goal: After this module you can design a minimal but rigorous experiment for your semina project — on paper, before writing a line of code.

Conditions, Controls, and Ablations

Definitions

Term	Meaning
Control	unchanged reference — the baseline
Condition	a variant differing in one factor
Ablation	full system *minus* one component

Example — RAG pipeline

- Baseline: BM25 + GPT-4
- Condition A: Dense retrieval + GPT-4
- Condition B: Dense + BM25 hybrid + GPT-4

Design rule: one variable at a time

If Condition B differs from A in *both* retrieval method *and* prompt format, you cannot isolate the cause of any difference.

> Change one thing at a time. When unavoidable, document the confound explicitly.

Minimum viable experiment

1. One baseline (simple, strong, cited)
2. One primary condition (your system)
3. One ablation (system minus your key component)
4. Two metrics (primary + secondary)
5. Fixed dataset split

- Ablation: B without re-ranking
- Ablation: B without query expansion

Each ablation isolates one contributing factor.

> Ablations are often more convincing than the main result — they show *what* drives the improvement.

Metrics and Evaluation

The core principle

The metric must directly measure the property your hypothesis claims. Derive the metric from the hypothesis — never the reverse.

Common metrics by task *(representative, not exhaustive)*

Task	Metric
Retrieval (ranked)	nDCG@k, MAP, Recall@k
Classification	F1 (macro/micro), MCC
Generation	ROUGE, BERTScore, BLEURT

Metric traps

Trap	Problem
BLEU for faithfulness	measures overlap, not factuality
Accuracy on imbalanced data	majority class trivially wins
Mean without variance	one lucky run inflates results
Proxy without validation	high proxy, low human preference

Dataset quality matters as much as the metric

Before using a benchmark, check:

- How was it constructed? By whom?
- Does it have known biases or artefacts?
- Are the labels reliable (inter-annotator agreement)?

Task	Metric
Faithfulness (RAG)	FactScore, Ragas faithfulness
Human eval	Likert, pairwise preference

> A biased dataset with a correct metric still produces a misleading result.

Common Pitfalls in ML Research

Data pitfalls

- **Train/test overlap** — shuffling before split leaks test items into training
- **Temporal leakage** — future data used to predict past events
- **Benchmark overfitting** — tuning on the test set, even unintentionally
- **Label noise** — gold labels wrong in 5–15% of NLP datasets
- **Dataset bias** — benchmark constructed from a specific source generalises poorly

Resource pitfalls

- **Insufficient compute** — full experiment requires GPU-hours not available

Model and reporting pitfalls

- **Spurious correlations** — model learns dataset artefacts, not the task
- **LLM-as-judge trained on same distribution** — evaluator is biased toward the model it judges
- **p-hacking** — try many metrics, report only the significant one
- **HARKing** — present post-hoc hypothesis as pre-specified
- **Selective ablations** — run 10, report 3
- **No error bars** — single run, no variance

Structural safeguards

1. Write hypothesis *before* seeing results
2. Commit evaluation script to Git *before* running on test set
3. Log every run — runs you don't report still happened

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- **Storage underestimated** — embeddings, checkpoints, logs exceed quota
- **API cost overrun** — LLM calls at scale cost 10× the estimate

> Estimate resources before running. A design that cannot be executed is not a design.

4. Report your worst result alongside your best

> Pre-registration turns pitfalls into choices — you have to actively decide to violate them.

Source: Pineau et al. 2021; Dodge et al. 2019

Statistical Significance and Effect Size

Statistical tests for ML

Situation	Test
Two systems, paired samples	Wilcoxon signed-rank
Two classifiers, same test set	McNemar's test
Means over multiple runs	Paired t-test ($n \geq 30$)
Multiple conditions vs. one baseline	Bonferroni correction

Threshold: $p < 0.05$ conventional, but:

- report the exact p-value
- $p = 0.049$ and $p = 0.051$ are not meaningfully different
- apply multiple-test correction when comparing many conditions

Effect size matters more than p

Measure	Use
Cohen's d	standardised mean difference
Δ metric (absolute)	nDCG +0.03 vs. +0.003
Relative improvement	"+5% over baseline"

Practical significance check

Even a significant +0.001 nDCG is rarely worth deploying. Ask:

- Is it reproducible on other datasets?
- Is it consistent across subgroups?
- Would a user notice the difference?

> Significant but tiny may matter less than non-significant but large and consistent.

Wrap-Up

The Research Loop

- 11 steps: Literature → Gap → Research Q. → Hypothesis → Design → Implement → Experiment → Results → Analyse → Write-up → Communicate → repeat
- Iteration is not failure — it is the mechanism of science
- Steps 1–5 determine quality more than steps 6–8

Experiment Design

- Conditions, controls, ablations: one variable at a time
- Metric follows from hypothesis — never the reverse

For your project — do this now:

1. Write your research question in one sentence (method · dataset · metric)
2. State a falsifiable, directional hypothesis
3. Name your baseline and justify it
4. Define your primary metric and why it matches your hypothesis
5. Estimate your resource budget (compute · data · time)

Next session (S03): literature search in depth, reference management, the publication process, project management with GTD and Git.

> Bring a written research question and hypothesis to S03. These will be reviewed as part of your

- Pre-register: design before you see results
- Estimate resources before you commit to a design

research narrative (due 12/05).

Image Credits & References

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